

Analytical problem solving Activity

1. Gradient Magnitude & Gradient direction

Given, $G_x = 24$ $G_y = 18$

(a) Gradient magnitude, $G = \sqrt{G_x^2 + G_y^2}$

$$= \sqrt{24^2 + 18^2} \Rightarrow \sqrt{576 + 324}$$

$$= \sqrt{900} \Rightarrow 30$$

(b) Gradient direction

$$\theta = \tan^{-1} \left(\frac{G_y}{G_x} \right)$$

$$= \tan^{-1} \left(\frac{18}{24} \right) \Rightarrow \tan^{-1} (0.75)$$

$$\theta \approx 36.87^\circ$$

2. Given,

Pixel	P1	P2	P3	P4	P5
Magnitude	25	50	90	60	30

$\Rightarrow$ Gradient direction is horizontal

It keeps a pixel only if it has a larger gradient value than its neighboring pixels in the gradient direction.

- $p_1 = 25 \rightarrow$ smaller than $p_2 = 50 \Rightarrow$ suppress
- $p_2 = 50 \rightarrow$ smaller than $p_3 = 90 \Rightarrow$ suppress
- $p_3 = 90 \rightarrow$ greater than $p_2 = 50$ and $p_4 = 60 \Rightarrow$ keep
- $p_4 = 60 \rightarrow$ smaller than $p_3 = 90 \Rightarrow$ suppress
- $p_5 = 30 \rightarrow$ smaller than $p_4 = 60 \Rightarrow$ suppress

After Non-Maximum Suppression:

<u>Pixel</u>	p_1	p_2	p_3	p_4	p_5
<u>Result</u>	0	0	90	0	0

$\therefore p_3$ is 90 and retained, all other pixels were suppressed.

3. Given,

High threshold = 100

Low threshold = 50

<u>pixels</u>	p_1	p_2	p_3	p_4	p_5	p_6
<u>Magnitude</u>	135	90	45	120	70	20

Rules

Magnitude $\geq 100 \rightarrow$ strong edge

Magnitude 50 to $< 100 \rightarrow$ weak edge

Magnitude $< 50 \rightarrow$ no edge

<u>Pixel</u>	p_1	p_2	p_3	p_4	p_5	p_6
<u>Magnitude</u>	135	90	45	120	70	20
<u>classification</u>	strong edge	weak edge	Non edge	strong edge	weak edge	Non edge

4.

<u>Pixel</u>	P_1	P_2	P_3	P_4	P_5	P_6
<u>Magnitude</u>	130	95	40	115	65	25

Threshold :

High threshold = 100

Low threshold = 50

<u>Pixel</u>	P_1	P_2	P_3	P_4	P_5	P_6
<u>Magnitude</u>	130	95	40	115	65	25
<u>Edge</u>	strong edge	strong edge	non-edge	strong edge	weak edge	non-edge

Applying hysteresis:

- P_2 is connected to P_1
- P_5 is not connected to any strong edge

 $\therefore P_1, P_2, P_4, P_7$ & P_5 is removed

1.

Given,

Edge points: $P_1(5, 4)$
 $P_2(7, 6)$ Candidate centers: $C_1(2, 4)$
 $C_2(4, 3)$

Radius from C_1 to P_1 : $r = \sqrt{(5-2)^2 + (4-4)^2}$
 $= \sqrt{9} \Rightarrow 3$

Radius from C_1 to P_2 : $r = \sqrt{(7-2)^2 + (6-4)^2} \Rightarrow \sqrt{5^2 + 2^2}$
 $= \sqrt{29} \approx 5.39$

Radius from C_2 to P_1 : $r = \sqrt{(5-4)^2 + (4-3)^2}$
 $= \sqrt{1^2 + 1^2} \Rightarrow \sqrt{2} \approx 1.41$

Radius from C_1 to P_2 : $r = \sqrt{(7-4)^2 + (6-3)^2}$
 $= \sqrt{3^2 + 3^2} \Rightarrow \sqrt{18} \approx 4.24$

$\therefore$ There is no common radius for the two edge points for either candidate center. So, neither center can be selected based on a matching-radius vote with given points.

2. Given,

Edge : $P(8,6)$ center $(4,3)$

$$r = \sqrt{(x-x_c)^2 + (y-y_c)^2}$$

$$= \sqrt{(8-4)^2 + (6-3)^2} \Rightarrow \sqrt{4^2 + 3^2} \Rightarrow \sqrt{25} = 5$$

$r = 5 \text{ units}$

3. Given,

$C = (5,4)$, $P = (8,8)$

$$r = \sqrt{(x-x_c)^2 + (y-y_c)^2}$$

$$= \sqrt{(8-5)^2 + (8-4)^2} \Rightarrow \sqrt{3^2 + 4^2} = \sqrt{25} = 5$$

$r = 5 \text{ units}$

